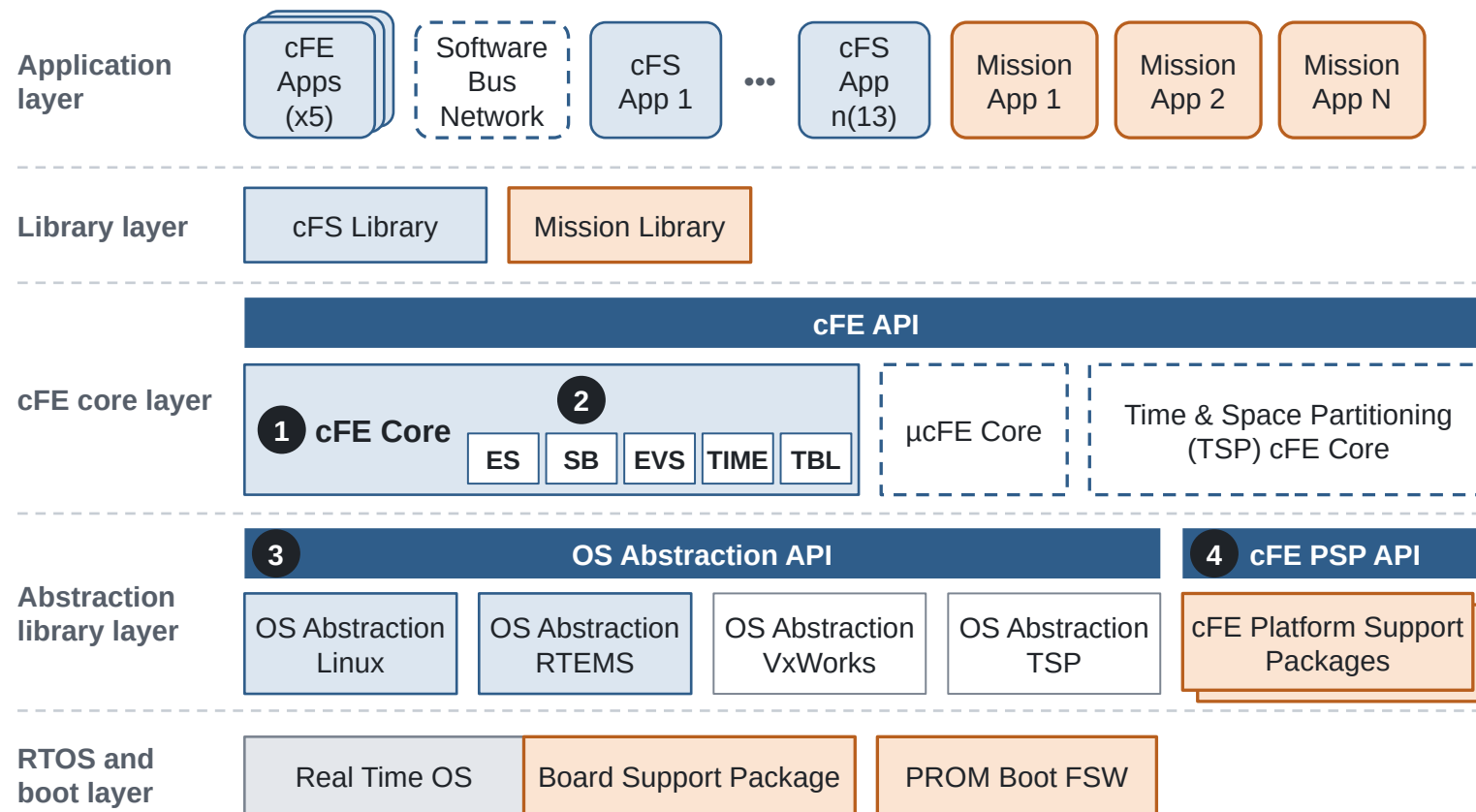


The OSAL and PSP hide the real-time OS and the board, so the cFE and the apps above it stay platform independent



- 1 cFE Core services** pp. 25, 45, paraphrase
Provide the services common to most flight apps: ES, SB, EVS, TIME and TBL
- 2 Software bus (SB)** p45
“Provides an application publish/subscribe messaging service”
- 3 OSAL (OS Abstraction API)** p23, paraphrase
One API to the cFE, whatever real-time OS runs underneath
- 4 PSP (cFE PSP API)** p24, paraphrase
One API to the avionics hardware and board support package

Status as of August 2015

Open source, NASA maintained	Mission developed
In development	Third party
No status color in source	Layer API

Source: diagram rebuilt from McComas et al., NASA Goddard, Aug. 2015, NTRS 20150018075, p8. Service acronyms and order: 2021 memo, printed p45 (the 2015 deck lists the same five services on p9). Roles: NASA/TM-20205000691/REV 1, Jan. 2021, NTRS 20205011588, printed page on each heading. Dark bars: each layer’s standard API (2023 overview, NTRS 20230002444, slide 8).